Examiner only

Option C: Neurobiology and Behaviour

9. Prairie dogs (*Cynomys ludovicianus*) are rodents, native to the grasslands of North America, which live in large colonies. Members of a family group interact through a variety of social behaviours such as kissing and grooming one another. **Image 9.1** shows prairie dogs sharing a greeting kiss.

Image 9.1

(a) Prairie dogs are said to demonstrate social behaviour.

(i) Define the term social behaviour. [1]

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Prairie dogs communicate with other members of the colony by making different sounding barks. When an intruder approaches, the first prairie dog gives a sharp bark, bobs up and down, barks again, and then plunges into a burrow. Other lookouts further away repeat this behaviour alerting other members of the colony along the route taken by the intruder.

(ii) The social behaviour of 'barking and bobbing' is an example of a fixed action pattern. Explain how such a fixed action pattern is generated and why it may be advantageous to the animals within the social group. [2]

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(iii) State **two** differences between a fixed action pattern and a reflex action. [2]

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(b) Social behaviour involves a large number of learnt responses.

Researchers investigated the behaviour of prairie dogs in colonies in Colorado. They were trying to find out if prairie dogs became habituated to the arrival of a human in their territory.

(i) Define the term habituation and explain why it is an advantage to the prairie dogs. [2]

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The researchers selected two colonies of prairie dogs:

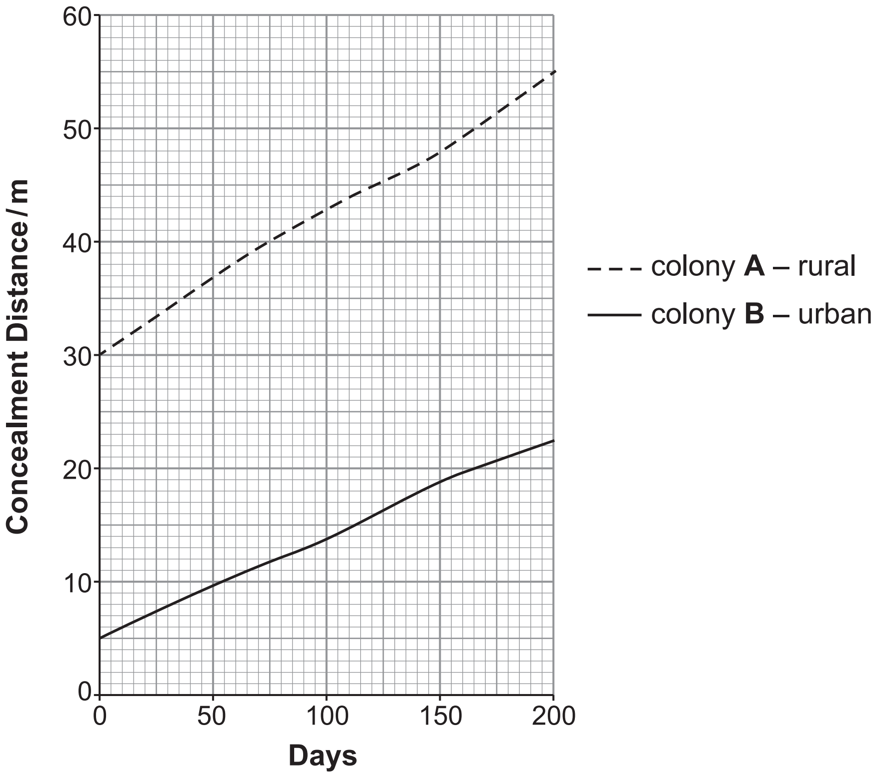
Colony **A** From a rural area, out of town, and not used to human presence.

Colony **B** From an urban area, close to city parks, where the prairie dogs would occasionally encounter humans.

Humans slowly approached each colony 100 times over 200 days. They recorded the concealment distance – this is the distance from the lookout animal to the human when the prairie dogs went down into the burrows. The researchers expected that the prairie dogs would become habituated to the presence of humans.

The results are shown in **Graph 9.2**.

Graph 9.2



(ii) Using **Graph 9.2**, calculate the percentage increase in concealment distance for the rural prairie dogs over the 200 days. [2]

Percentage increase in concealment distance = ...

(iii) Use the evidence from **Graph 9.2** to conclude whether the researchers' expectations were correct. [3]

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(c) **Images 9.3** and **9.4** show fMRI scans of two patients taken whilst carrying out language skill tasks.

Image 9.3

fMRI taken from a healthy adult patient

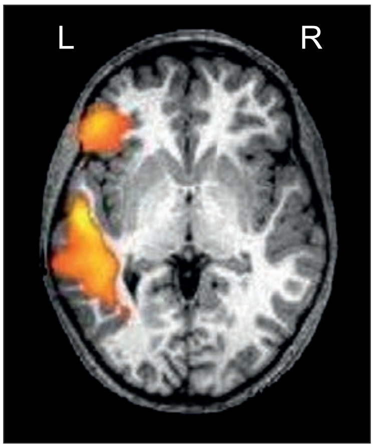
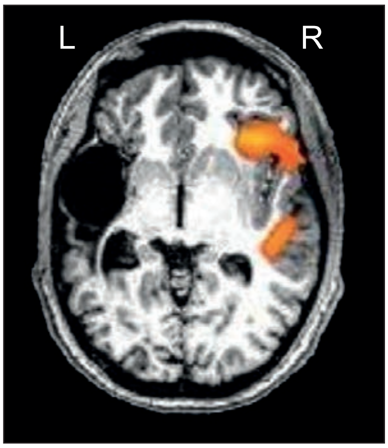


Image 9.4

fMRI taken from an adult patient who had a stroke shortly after birth



(i) State why an fMRI scan was the most suitable form of brain scan to show where the patients were processing language. [1]

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(ii) Suggest the area of the brain where the stroke took place and explain what has happened in the brain of the stroke patient in **Image 9.4** to account for the fMRI scan result. [4]

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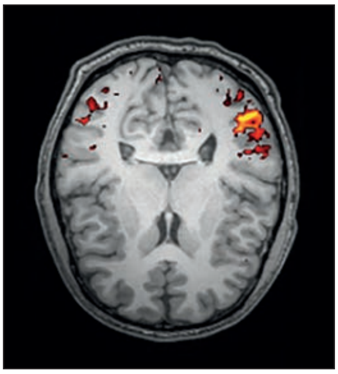
Examiner only

The patient in **Image 9.4** regained all aspects of language. **Image 9.5** shows an fMRI scan from a patient who had a stroke as a mature adult.

Image 9.5



(iii) Using **Image 9.5** and your own knowledge, explain why it would be more difficult for this patient to fully regain all aspects of language. [3]

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END OF PAPER